

DD2424 Assignment 4 Simon Yu

Simon Yu

May 2026

1 Analytical Gradient Check

As in previous assignments the analytical gradient was checked against the PyTorch calculated gradient. As can be seen in the results the errors all have a very small order and the gradients hold the same shape thus we can say that our analytical gradient computation is correct.

Comparing W:

```
shape my_grads    : (10, 10)
shape torch_grads: (10, 10)
max absolute difference: 1.9081958235744878e-17
max relative error   : 1.5539745330827644e-14
```

Comparing U:

```
shape my_grads    : (10, 83)
shape torch_grads: (10, 83)
max absolute difference: 4.336808689942018e-17
max relative error     : 1.2299768611627778e-14
```

Comparing b:

```
shape my_grads    : (10, 1)
shape torch_grads: (10, 1)
max absolute difference: 8.326672684688674e-17
max relative error     : 1.9741670599493563e-15
```

Comparing V:

```
shape my_grads    : (83, 10)
shape torch_grads: (83, 10)
max absolute difference: 6.938893903907228e-18
max relative error     : 2.3978840974272957e-13
```

Comparing c:

```
shape my_grads    : (83, 1)
shape torch_grads: (83, 1)
max absolute difference: 2.7755575615628914e-17
```

max relative error : 2.0467962999938244e-16

2 Loss function

After 133000 update steps (~ 3 epochs) a smooth loss of 1.5533 was achieved, this can be seen in Figure 1

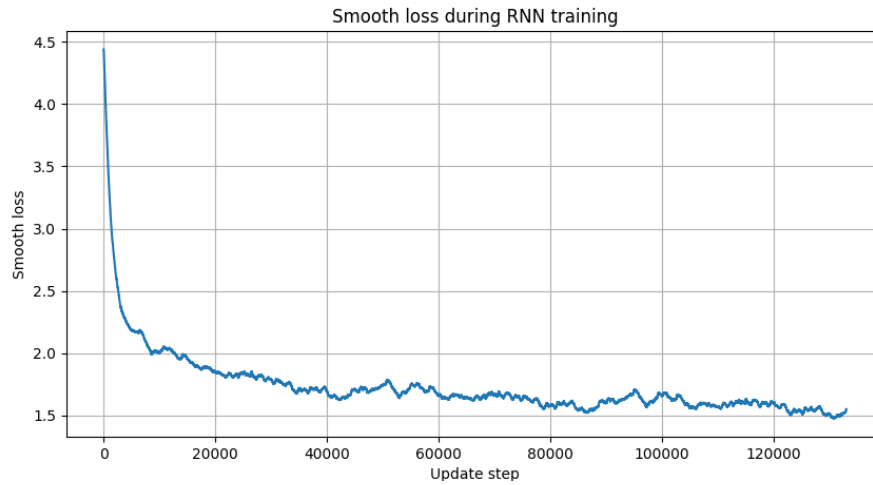


Figure 1: Smooth loss function over 133000 update steps (~ 3 epochs)

3 Synthesized text

```
-----
Sample at iteration 1
-----
-      f(20
(06iY?uvHh/0'I  j!M4}AFc 1DsorPMdjruB ;u93VF4C 1lLvge!07
  RhK;YkekCRj!r"Bu kR9le3xG(4TSP^;Dm0/wD; VXB }Hht_q!
    eZuItHCQFcZ'"S(PV_n7HHcy61d-q Je J0zpdCe vBU6.
      Ji0HnESQVxS2q  TPZ6Ri^-tUL0qk .xF.;i PqD}u6
-----

iter = 10000, smooth_loss = 2.0127622846455857
-----
Sample at iteration 10000
-----
, bre ke DitWers. "He's aid diTk, of me picrking butt an sha
  bremars focharl asle and the ereitily, "Every un
  Hermiane." waif Hrolldot," said chomhartedl, becate them
  gromide t an. Amen th ig th Dabl
```

```

-----
iter = 20000, smooth_loss = 1.8466537068864128
-----
Sample at iteration 20000
-----
uscrester. . ."
"Ostlaking waid I'ving to eving an Sreacel cont wele mean
    levgots avaurik a fout rough whair aboutthat he dieny the
    Snotelly theme atoce. Harry was s. here ich Maxime sco-e
    . Hagrid, Ha-----

iter = 30000, smooth_loss = 1.7848902258857349
-----
Sample at iteration 30000
-----
ward murmersuse wit igsor that sied, th mace abous om he
    furely through largive. He doed and wkide of Thee't donk
    sot nofl..... .Y.
"We're air serrendorserors that wo fol wite at him youble c
-----

iter = 40000, smooth_loss = 1.6834399281106074
-----
Sample at iteration 40000
-----
it could dimess thrye of the the gut. He "Muthren shouse he
    that interens - and allted ladger in they bellise,
    sagoner himbledditing Vordess of w; nee right'n whother
    wopsing the grown that pringng h
-----

iter = 50000, smooth_loss = 1.7415933207751617
-----
Sample at iteration 50000
-----
ext introut! They weond poiat en it.
"Weath homem nightlis happeited the tilroy, and he dong they
    ham whien, the pouls veit of in chasctitchably of
    thehrit, Ron mango everything they wency and would
-----

iter = 60000, smooth_loss = 1.6632148381372138
-----
Sample at iteration 60000
-----
auon at hadrned the Greatwert punk was id header," said
    Renered see plairing applane Harry, "for the renfimated
    tur grow sagging the leys it to nelt realing as Half
    Hermione once the? Their eevering

```

```

-----
iter = 70000, smooth_loss = 1.6674227879891241
-----
Sample at iteration 70000
-----
Babooth?"
"No! "Ked," said Hermione. Halred Spatchising to patce, you
    cansto deempeching," said Ron rate. "I dabbearely sm!"
    they were suspded. "It low-though they were coulds of
    Vory chand of buncer
-----

iter = 80000, smooth_loss = 1.5894346369569452
-----
Sample at iteration 80000
-----
his gelr."
He with in to jack. He withing in his eround moont off
    Siryure aloud Hoorvess was a deed bobss. The. He
    ofesled," said as Hermiome malkers what was fuch
    infornstrote molfs were would nog
-----

iter = 90000, smooth_loss = 1.603305849606402
-----
Sample at iteration 90000
-----
ntinded, and the mack trememalling lillid be ager-inness
    flewnew him hich saids staying he holdever to fylit him
    stile pore the limplied to dow, with erhing to he'd inkay
    loavisiftares. But it was al
-----

iter = 100000, smooth_loss = 1.6521650608242602
-----
Sample at iteration 100000
-----
    said Bagments this might and see weres. "So head earun.
    "Cound to thurly, who walleble. "Ly gaiss frieh sayelizariud
        be with over they's ormitche, "the can't tere. "
        Dumbledore? Then who in tean," sa
-----

```

4 Best model

After training the model for 200000 update step the model with the lowest loss synthesized this 1000 character passage:

e was goody bedy
 " should this entered the chay, don't sence.
 Host the ungreles - one of up at could betwlety," said
 Mrsfatees are arouh ussed is what Ain are are as thes
 officinip sligg timan viement.

Therice, again. Harry Madam Potagasting about it, and
 Hermarivacted outiog his his goar carroushing will reguid
 . His still was not he wivere?"
 "That's old reture - get and a delbed Mossorfush. "Com the
 hampt leve.
 "Yes upostion Sirius. She badget, inders sament; her hure
 his surs dest bed around Ministed the the faiting perible
 .
 "What drelly ..."
 Here he wath his he could head wand, caught and called
 luquid," he was sance; the Ropening. propled arounder
 around. She sant of and continubled orgering that yeal
 the momently behams. Mrsakel would mother was?"
 "No, and Sperely seemes Mrs.oW and Mrs, a turg the hosparily
 of. What's sutting an art in yor, again by and could
 you opening.
 MuA Magate the preguodicary brievard him beborins ligk
 Dumbledore lain?
 "It's giver ince. . .". . .